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Light Emitting Display Device Including Hydrogen or Oxygen Absorbing Layer

Abstract

A light emitting display device includes a lower substrate, a thin film transistor on the lower substrate, a passivation layer disposed on the thin film transistor and including hydrogen, an overcoating layer disposed on the passivation layer and planarizing the passivation layer, a light emitting element disposed on the overcoating layer and including an anode, a light emitting layer on the anode, and a cathode on the light emitting layer, a bank disposed on the overcoating layer and defining a light emitting area, an adhesive layer on the light emitting element and the bank, and a hydrogen absorbing layer disposed on the adhesive layer and including a hydrogen absorbing filler, wherein a side end of the bank is disposed more inwardly than side ends of the adhesive layer and the hydrogen absorbing layer, wherein the side ends of the adhesive layer and the hydrogen absorbing layer are disposed more inwardly than a side end of the overcoating layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/197,406, filed on May 15, 2023, which is a continuation of U.S. patent application Ser. No. 17/137,897, filed on Dec. 30, 2020, now U.S. Pat. No. 11,690,248, issued on Jun. 27, 2023, which claims the priority benefit of Korean Patent Application No. 10-2019-0179939 filed on Dec. 31, 2019, and Korean Patent Application No. 10-2020-0076718 filed on Jun. 23, 2020, which are hereby incorporated by reference in their entirety for all purposes as if fully set forth herein.

BACKGROUND

Field of the Disclosure

[0002] The present disclosure relates to a light emitting display device and more particularly, to a light emitting display device capable of reducing defects of a thin film transistor due to hydrogen.

Description of the Background

[0003] A light emitting display device is a self-light emitting display device, and may be manufactured to be light and thin since it does not require a separate light source, unlike a liquid crystal display device having a separate light source. In addition, the light emitting display device has advantages in terms of power consumption due to a low voltage driving, and is excellent in terms of a color implementation, a response speed, a viewing angle, and a contrast ratio (CR). Therefore, light emitting display devices are being studied as next-generation display devices.

[0004] Such a light emitting display device emits light by driving a light emitting element disposed in each sub-pixel. At this time, one or more thin film transistors (TFT) electrically connected to the light emitting element are disposed in each sub-pixel to independently drive the light emitting element of each sub-pixel.

SUMMARY

[0005] A thin film transistor includes a gate electrode, a source electrode, a drain electrode, and a semiconductor layer. The source electrode and the drain electrode contact the semiconductor layer, and the gate electrode is disposed to overlap the semiconductor layer. When a gate voltage more than a predetermined voltage is applied to the gate electrode of the thin film transistor, a channel is formed in the semiconductor layer so that current flows between the source electrode and the drain electrode. As such, the thin film transistor has switching characteristics, and such switching characteristics may be determined by various factors. For example, when a material of the semiconductor layer is deformed, mobility of the thin film transistor may be changed, and the switching characteristics of the thin film transistor may be changed.

[0006] A light emitting display device includes a passivation layer for protecting the light emitting element from moisture or oxygen which penetrates the light emitting element. The passivation layer is formed on an optical element to protect the light emitting element.

[0007] Such a passivation layer is formed by for example, chemical vapor deposition (CVD) using silane (SiH_4) and ammonia (NH_3). A small amount of hydrogen may be generated from silane and ammonia in a film-formation process of the passivation layer by chemical vapor deposition. Accordingly, hydrogen generated in the film-formation process of the passivation layer may be diffused into and included in the passivation layer. Residual hydrogen included in the passivation layer may move inside the light emitting element. When the residual hydrogen diffuses into and reacts with the semiconductor layer of the thin film transistor, characteristics of the thin film transistor may be changed. Accordingly, hydrogen generated in the film-formation process of the passivation layer may degrade performance of the light emitting display device as well as performance of the thin film transistor.

[0008] To solve a problem in which hydrogen generated when the passivation layer of the light emitting display device is formed remains in the light emitting display device and as the remaining hydrogen diffuses to the thin film transistor, characteristics of the thin film transistor are changed to thereby degrade the performance of the light emitting display device, as described above, the inventors of the present disclosure have invented a novel structure of a light emitting display device capable of absorbing hydrogen included in a passivation layer, and a method of manufacturing the same.

[0009] Accordingly, the present disclosure provides a light emitting display device capable of absorbing hydrogen remaining in a passivation layer and preventing performance of a thin film transistor from being degraded.

[0010] In addition, the present disclosure provides a light emitting display device having a structure capable of removing hydrogen in a passivation layer without additional equipment for removing the hydrogen generated in a film-formation process of the passivation layer.

[0011] Also, the present disclosure can significantly improve a performance of removing hydrogen inside a light emitting display device without lowering durability or light emission characteristics of the light emitting display device.

[0012] Further, the present disclosure provides a light emitting display device having a structure capable of blocking oxygen injected to the inside from the outside thereof.

[0013] Furthermore, the present disclosure can solve defects of white or dark spots of pixels generated by hydrogen or oxygen.

[0014] The present disclosure is not limited to the above-mentioned objects, and other objects, which are not mentioned above, can be clearly understood by those skilled in the art from the following descriptions.

[0015] According to an exemplary aspect of the present disclosure, a light emitting display device includes a lower substrate; a thin film transistor on the lower substrate; a passivation layer disposed on the thin film transistor and including hydrogen; an overcoating layer disposed on the passivation layer and planarizing the passivation layer; a light emitting element disposed on the overcoating layer and including an anode, a light emitting layer on the anode, and a cathode on the light

emitting layer; a bank disposed on the overcoating layer and defining a light emitting area; an adhesive layer on the light emitting element and the bank; and a hydrogen absorbing layer disposed on the adhesive layer and including a hydrogen absorbing filler, wherein a side end of the bank is disposed more inwardly than side ends of the adhesive layer and the hydrogen absorbing layer, wherein the side ends of the adhesive layer and the hydrogen absorbing layer are disposed more inwardly than a side end of the overcoating layer.

[0016] According to another aspect of the present disclosure, a light emitting display device includes a lower substrate; a thin film transistor on the lower substrate; a passivation layer disposed on the thin film transistor and including hydrogen; an overcoating layer disposed on the passivation layer and planarizing the passivation layer; a light emitting element disposed on the overcoating layer and including an anode, a light emitting layer on the anode, and a cathode on the light emitting layer; a bank disposed on the overcoating layer and defining a light emitting area; and an adhesive layer disposed on the light emitting element and the bank and including a hydrogen absorbing filler or an oxygen absorbing filler, wherein the hydrogen absorbing filler or the oxygen absorbing filler is disposed to be adjacent to the passivation layer.

[0017] Other detailed matters of the exemplary aspects are included in the detailed description and the drawings.

[0018] According to the present disclosure, it is feasible to prevent a reduction in performance of a thin film transistor by configuring a hydrogen absorbing layer capable of absorbing residual hydrogen in a passivation layer, which may change characteristics of the thin film transistor.

[0019] According to the present disclosure, damage to a light emitting element due to moisture or foreign matter can be prevented by further including an upper protective layer for protecting the light emitting element, so that element reliability can be greatly improved.

[0020] The effects according to the present disclosure are not limited to the contents exemplified above, and more various effects are included in the present specification.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The above and other aspects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0022] FIG. 1 is a plan view of a light emitting display device according to an exemplary aspect of the present disclosure;

[0023] FIG. 2 is a cross-sectional view of one sub-pixel of the light emitting display device according to an exemplary aspect of the present disclosure;

[0024] FIG. 3 is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure;

[0025] FIG. 4 is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure;

[0026] FIG. 5 is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure;

[0027] FIG. 6 is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure;

[0028] FIG. 7 is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure;

[0029] FIG. 8 is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure; and

[0030] FIG. 9 is a cross-sectional view of one sub-pixel of a light emitting display device according

to still another exemplary aspect of the present disclosure.

DETAILED DESCRIPTION

[0031] Advantages and characteristics of the present disclosure and a method of achieving the advantages and characteristics will be clear by referring to exemplary aspects described below in detail together with the accompanying drawings. However, the present disclosure is not limited to the exemplary aspects disclosed herein but will be implemented in various forms. The exemplary aspects are provided by way of example only so that those skilled in the art can fully understand the disclosures of the present disclosure and the scope of the present disclosure. Therefore, the present disclosure will be defined only by the scope of the appended claims.

[0032] The shapes, sizes, ratios, angles, numbers, and the like illustrated in the accompanying drawings for describing the exemplary aspects of the present disclosure are merely examples, and the present disclosure is not limited thereto. Like reference numerals generally denote like elements throughout the specification. Further, in the following description of the present disclosure, a detailed explanation of known related technologies may be omitted to avoid unnecessarily obscuring the subject matter of the present disclosure. The terms such as “including,” “having,” and “consist of” used herein are generally intended to allow other components to be added unless the terms are used with the term “only”. Any references to singular may include plural unless expressly stated otherwise.

[0033] Components are interpreted to include an ordinary error range even if not expressly stated.

[0034] When the position relation between two parts is described using the terms such as “on”, “above”, “below”, and “next”, one or more parts may be positioned between the two parts unless the terms are used with the term “immediately” or “directly”.

[0035] When an element or layer is disposed “on” another element or layer, another layer or another element may be interposed directly on the other element or therebetween.

[0036] Although the terms “first”, “second”, and the like are used for describing various components, these components are not confined by these terms. These terms are merely used for distinguishing one component from the other components. Therefore, a first component to be mentioned below may be a second component in a technical concept of the present disclosure.

[0037] Like reference numerals generally denote like elements throughout the specification.

[0038] A size and a thickness of each component illustrated in the drawing are illustrated for convenience of description, and the present disclosure is not limited to the size and the thickness of the component illustrated.

[0039] The features of various aspects of the present disclosure can be partially or entirely adhered to or combined with each other and can be interlocked and operated in technically various ways, and the aspects can be carried out independently of or in association with each other.

[0040] Hereinafter, a light emitting display device according to exemplary aspects of the present disclosure will be described in detail with reference to accompanying drawings.

[0041] FIG. 1 is a plan view of a light emitting display device according to an exemplary aspect of the present disclosure.

[0042] FIG. 2 is a cross-sectional view of one sub-pixel of the light emitting display device according to an exemplary aspect of the present disclosure.

[0043] Specifically, FIG. 2 is a cross-sectional view of one side of a light emitting display device **100**.

[0044] With reference to FIGS. 1 and 2, the light emitting display device **100** includes a lower substrate **110**, a gate insulating layer **120**, a thin film transistor TFT, a passivation layer **130**, a light emitting element **140**, an overcoating layer **150**, a bank **160**, an adhesive layer **170**, a hydrogen absorbing layer **180**, and an upper substrate **119**.

[0045] For reference, FIG. 1 illustrates only the lower substrate **110** and the upper substrate **119** of the light emitting display device **100** for convenience of description.

[0046] With reference to FIG. 1, the light emitting display device **100** includes a plurality of pixels

PX. Each of the pixels PX includes a plurality of sub-pixels SP. The sub-pixels SP may be basic light emitting units constituting the light emitting display device **100**, and the respective sub-pixels SP each constituting one pixel PX emit light of different colors. As illustrated in FIG. **1**, the sub-pixels SP may include a red sub-pixel, a green sub-pixel, a blue sub-pixel, and a white sub-pixel, or may include only a red sub-pixel, a green sub-pixel, and a blue sub-pixel.

[0047] In addition, with reference to FIG. **2**, the light emitting display device **100** according to an exemplary aspect may include sub-pixels SP having a color-filter on transistor (COT) structure. For example, a color of each sub-pixel SP may be implemented by a color filter CF disposed on an upper layer of the thin film transistor TFT so as to overlap a light emitting area of the sub-pixel SP. For example, when the light emitting element **140** emits white light, a red color filter CF is disposed in an area corresponding to a light emitting area of the light emitting element **140**, whereby a red sub-pixel SP may be implemented. In addition, a green color filter CF is disposed in an area corresponding to the light emitting area of the light emitting element **140**, whereby a green sub-pixel SP may be implemented. Also, a blue color filter CF is disposed in an area corresponding to the light emitting area of the light emitting element **140**, whereby a blue sub-pixel SP may be implemented. Further, a white sub-pixel SP may be implemented by not disposing the color filter CF in an area corresponding to the light emitting area of the light emitting element **140**. Concrete contents of the color filter CF will be described later.

[0048] The thin film transistor TFT including a semiconductor layer A, a gate electrode G, a source electrode S, and a drain electrode D is formed on the lower substrate **110** formed of an insulating material. For example, the semiconductor layer A is formed on the lower substrate **110**, and the gate insulating layer **120** for insulating the gate electrode G and the semiconductor layer A is formed on the semiconductor layer A and the lower substrate **110**. The gate electrode G is formed on the gate insulating layer **120**, and the source electrode S and the drain electrode D are formed on the semiconductor layer A and the gate insulating layer **120**. The source electrode S and the drain electrode D are electrically connected to the semiconductor layer A in a manner of being in contact with the semiconductor layer A, and are formed on some areas of the gate insulating layer **120**. For convenience of description, only a driving thin film transistor among various thin film transistors that may be included in the light emitting display device **100** is illustrated, but a switching thin film transistor may also be included. In addition, in the specification, although the thin film transistor TFT is described as having a coplanar structure, a thin film transistor having an inverted staggered structure may also be used.

[0049] The semiconductor layer A may be formed of an oxide semiconductor or an amorphous semiconductor. The amorphous semiconductor may be formed of amorphous silicon. As another example, the semiconductor layer A may be formed of polycrystalline silicon that can be applied to the driving thin film transistor within the pixel since polycrystalline silicon has mobility superior than that of amorphous silicon and has low energy consumption and high reliability. However, the present disclosure is not limited thereto.

[0050] The passivation layer **130** is conformally disposed on an upper surface of the thin film transistor TFT. For example, the passivation layer **130** may be formed along a shape of the upper surface of the thin film transistor TFT. In addition, the passivation layer **130** may be formed to cover the thin film transistor TFT, and the passivation layer **130** may be disposed on a front surface of the lower substrate **110** on which the thin film transistor TFT is disposed. Further, with reference to FIG. **1**, the passivation layer **130** is formed to overlap all of the plurality of sub-pixels SP.

[0051] In addition, the passivation layer **130** may be formed to have a thickness of 10 μm or less, for example, a thickness of 0.5 μm , but is not limited thereto.

[0052] As described above, the passivation layer **130** may be disposed to cover the thin film transistor TFT. The passivation layer **130** may protect the thin film transistor TFT from oxygen or moisture from the outside of the light emitting display device **100**. Various organic or inorganic films may be used for the passivation layer **130**, and the passivation layer **130** may be formed of

various structures such as an organic film-alone deposition structure, an inorganic film-alone deposition structure, or a structure in which an organic film and an inorganic film are alternately disposed. The passivation layer **130** may be formed of, for example, silicon nitride (SiNx), silicon oxide (SiOx) or the like, but is not limited thereto.

[0053] The color filter CF may be disposed on the passivation layer **130** so as to overlap the light emitting area of the sub-pixel SP. In addition, since the color filter CF is disposed in the light emitting area of the sub-pixel SP, it is disposed on an upper layer of the thin film transistor TFT, but is disposed not to overlap the thin film transistor TFT.

[0054] In addition, each color filter CF may include a red color filter CF including a red pigment, a green color filter CF including a green pigment, and a blue color filter including a blue pigment to implement a color of the sub-pixel SP.

[0055] Also, to form the color filter CF, each of the red color filter CF, the green color filter CF, and the blue color filter CF needs to undergo processes such as a coating process, an exposure process, a development process, a post-bake process, and the like.

[0056] Among these processes, in the post-bake process, since a pigment layer for forming the color filter CF is heated, gases including hydrogen included in the pigment layer are discharged.

[0057] Despite of the post-bake process, hydrogen may remain in the color filter CF. Therefore, residual hydrogen included in the color filter CF can move inside the light emitting display device **100**, and the residual hydrogen diffuses to the semiconductor layer A of the thin film transistor TFT, whereby characteristics of the thin film transistor TFT can be changed. For example, the semiconductor layer A may absorb hydrogen. When the semiconductor layer A absorbs hydrogen, a threshold voltage V_{th} of the thin film transistor TFT may be shifted to thereby increase mobility of the thin film transistor TFT. Accordingly, the characteristics of the thin film transistor TFT may be changed to thereby lower performance thereof, and defects such as the occurrence of speckles or bright spots and uneven luminance may occur. Accordingly, since hydrogen generated in a film-formation process of the color filter CF may degrade performance of the light emitting display device **100** as well as the performance of the thin film transistor TFT, it is necessary to remove residual hydrogen in the color filter CF.

[0058] The overcoating layer **150** is disposed on the gate insulating layer **120**, the thin film transistor TFT, the passivation layer **130**, and the color filter CF. For example, the overcoating layer **150** may be disposed on the front surface of the lower substrate **110** on which the gate insulating layer **120** and the passivation layer **130** are formed. More specifically, as illustrated in FIG. 2, one side end of the overcoating layer **150** may be disposed more inwardly than a side end of the lower substrate **110**. However, the present disclosure is not limited thereto, and the side end of the overcoating layer **150** and the side end of the lower substrate **110** may be collinearly disposed.

[0059] In addition, the overcoating layer **150** may be formed to cover the thin film transistor TFT and the passivation layer **130**, and may planarize upper portions of the thin film transistor TFT and the passivation layer **130**. The overcoating layer **150** may be formed of an insulating material.

[0060] The light emitting element **140** including an anode **141**, a light emitting layer **142**, and a cathode **143**, and the bank **160** are formed on the overcoating layer **150**. For example, the anode **141** for supplying holes to the light emitting layer **142** is formed on an upper surface of the overcoating layer **150**, the light emitting layer **142** is formed on the anode **141**, and the cathode **143** for supplying electrons to the light emitting layer **142** is formed on the light emitting layer **142**.

[0061] When the light emitting display device **100** is a bottom emission type, the anode **141** may include, for example, a transparent conductive layer formed of a transparent conductive oxide (Hereinafter, referred to as TCO), such as indium tin oxide (ITO), indium zinc oxide (IZO), ZnO or the like. In addition, the cathode **143** may be formed of a metallic material having low light transmittance.

[0062] When the light emitting display device **100** is a top emission type, the anode **141** may include a transparent conductive layer formed of a reflective layer and a transparent conductive

oxide on the reflective layer. In addition, the cathode **143** may be formed by thinly forming a metallic material having a low work function, or may be formed of a transparent conductive oxide. When the cathode **143** is formed of a transparent conductive oxide, multiple layers may be disposed between the cathode **143** and the light emitting layer **142** so that electrons can move through the cathode **143**. For example, a metal-doped layer may be disposed between the light emitting layer **142** and the cathode **143**. Also, an organic buffer layer may be additionally disposed between the light emitting layer **142** and the metal-doped layer.

[0063] The bank **160** may be formed to cover a side portion of the anode **141** and may define the light emitting area of each sub-pixel SP.

[0064] For example, as illustrated in FIG. 2, the bank **160** may be formed to contact the overcoating layer **150** and cover both side portions of the anode **141**. Accordingly, the bank **160** may block diffusion of moisture and impurities that may be introduced from the overcoating layer **150**.

[0065] In addition, when the light emitting display device **100** is a top emission type, the bank **160** overlaps the thin film transistor TFT so that the thin film transistor TFT is not exposed to the outside. As a result, leakage photocurrent due to external light is not generated in the thin film transistor TFT and thus, the thin film transistor TFT can apply an appropriate driving current to the light emitting element **140**.

[0066] In addition, the bank **160** may extend adjacent to the side end of the overcoating layer **150**. For example, a side end of the bank **160** may be closer to the side end of the overcoating layer **150** than a boundary of the light emitting area of the sub-pixel SP. As illustrated in FIG. 2, the side end of the bank **160** may be disposed more inwardly than the side end of the overcoating layer **150**.

[0067] In addition, the light emitting layer **142** and the cathode **143** may be disposed on the bank **160**. In more detail, the light emitting layer **142** may be extended and disposed on the bank **160**, and the cathode **143** may be disposed to cover an upper surface and a side surface of the light emitting layer **142** and an upper surface of the bank **160**. However, as illustrated in FIG. 2, the light emitting layer **142** is not limited to extending only to the upper surface of the bank **160**, but may also extend to a side surface of the bank **160**.

[0068] In addition, the adhesive layer **170** may be formed to cover the light emitting element **140**, the bank **160**, and the overcoating layer **150**. More specifically, the adhesive layer **170** may be disposed to cover an upper surface and a side surface of the cathode **143**, the upper surface and the side surface of the bank **160**, and only an upper surface of the overcoating layer **150**. For example, the adhesive layer **170** may cover both an outer surface of the light emitting element **140** and an outer surface of the bank **160**, but may not cover all of an outer surface of the overcoating layer **150**.

[0069] In other words, a side end of the adhesive layer **170** may be disposed more inwardly than the side end of the overcoating layer **150**, but may be disposed more outwardly than the side end of the bank **160**. For example, the side end of the overcoating layer **150**, the side end of the adhesive layer **170**, and the side end of the bank **160** may be sequentially disposed based on an outside of the light emitting display device **100**. More specifically, the side end of the bank **160** is disposed more inwardly than the side end of the adhesive layer **170**, and the side end of the adhesive layer **170** is disposed more inwardly than the side end of the overcoating layer **150**.

[0070] And, the adhesive layer **170** bonds the lower substrate **110** and the upper substrate **119** facing the lower substrate **110**. Accordingly, the adhesive layer **170** may include an adhesive material. For example, the adhesive layer **170** may be formed of an adhesive material in a liquid form or a film form. For example, the adhesive layer **170** may be formed of a resin, epoxy, or acrylic material, but is not limited thereto.

[0071] Meanwhile, the passivation layer **130** is formed by chemical vapor deposition (CVD) using gases containing hydrogen (example, silane (SiH₄) and ammonia (NH₃)). Hydrogen may be generated in a film-formation process of the passivation layer **130**, and the generated hydrogen may

be diffused into the passivation layer **130** and included in the passivation layer **130**. Here, the hydrogen may include hydrogen atoms (H) and hydrogen molecules (H₂). For example, hydrogen remaining inside the passivation layer **130** may be diffused in a driving direction, and diffusion may be performed according to a concentration gradient, for example, chemical potential. Therefore, residual hydrogen included in the passivation layer **130** can move inside the light emitting display device **100**, and the residual hydrogen diffuses to the semiconductor layer A of the thin film transistor TFT, thereby change characteristics of the thin film transistor TFT. For example, the semiconductor layer A may absorb hydrogen. When the semiconductor layer A absorbs hydrogen, the threshold voltage V_{th} of the thin film transistor TFT is shifted, whereby mobility of the thin film transistor TFT may increase. Accordingly, the characteristics of the thin film transistor TFT may be changed to thereby degrade performance thereof, and defects such as the occurrence of speckles or bright spots and uneven luminance may occur. Accordingly, since hydrogen generated in the film-formation process of the passivation layer **130** may degrade the performance of the light emitting display device **100** as well as the performance of the thin film transistor TFT, it is necessary to remove residual hydrogen in the passivation layer **130**. In addition, the characteristics of the thin film transistor TFT may be changed due to the residual hydrogen in the passivation layer **130** according to a material of the upper substrate **119**, so that bright spots may occur.

[0072] Accordingly, the hydrogen absorbing layer **180** may be disposed on the adhesive layer **170** in order to remove residual hydrogen included in the color filter CF and the passivation layer **130**. For example, a side end of the hydrogen absorbing layer **180** and the side end of the adhesive layer **170** may be collinearly disposed.

[0073] Thus, the side end of the hydrogen absorbing layer **180** may also be disposed more inwardly than the side end of the overcoating layer **150**, but may be disposed more outwardly than the side end of the bank **160**. For example, the side end of the overcoating layer **150**, the side ends of the adhesive layer **170** and the hydrogen absorbing layer **180**, and the side end of the bank **160** may be sequentially disposed based on the outside of the light emitting display device **100**. More specifically, the side end of the bank **160** is disposed more inwardly than the side ends of the adhesive layer **170** and the hydrogen absorbing layer **180**, and the side ends of the adhesive layer **170** and the hydrogen absorbing layer **180** are disposed more inwardly than the side end of the overcoating layer **150**. For example, since the overcoating layer **150** is disposed to cover all of the thin film transistor TFT, the gate insulating layer **120**, and the passivation layer **130**, diffusion of hydrogen by the passivation layer **130** may be prevented, and the diffused hydrogen may be absorbed by the hydrogen absorbing layer **180**.

[0074] In addition, the hydrogen absorbing layer **180** is formed to have a thickness of 5000 angstroms (Å) or less and planarizes the upper surfaces of the overcoating layer **150**, the bank **160**, and the light emitting element **140**. For example, the hydrogen absorbing layer **180** may be formed to have a maximum thickness of 50 μm, but is not limited thereto. In addition, the hydrogen absorbing layer **180** may be a sealing material, but is not limited to this term. The sealing material may be disposed on a front surface of the light emitting element.

[0075] The sealing material may be formed based on a curable resin. For example, the curable resin may be formed of a photocurable resin or a thermosetting resin. More specifically, the curable resin may be an epoxy resin or a polyolefin resin, but is not limited thereto.

[0076] In addition, a side seal member may be further included on one side surface of the overcoating layer **150**, one side surfaces of the adhesive layer **170** and the hydrogen absorbing layer **180**, and one side surface of the upper substrate **119**. The side seal member borders a side portion of the light emitting display device **100**. For example, the side seal member may contact one side surface of the overcoating layer **150**, one side surfaces of the adhesive layer **170** and the hydrogen absorbing layer **180**, and one side surface of the upper substrate **119**. And, the side seal member can minimize moisture permeation in the side portion. To this end, the side seal member

may be formed of, for example, a curable material such as an acrylic curable material, a urethane curable material, and a silicon curable material, but is not limited thereto.

[0077] The hydrogen absorbing layer **180** described above may include a hydrogen absorbing filler that serves as a filter by adsorbing or absorbing hydrogen, for example, hydrogen atoms (H) and hydrogen molecules (H₂), or moisture, in order to absorb residual hydrogen in the color filter CF and/or the passivation layer **130**. The filler may be a powder, but is not limited to this term. For example, the hydrogen absorbing filler can be an inorganic material or an organic material.

[0078] As the hydrogen absorbing filler, the inorganic material may include one or more of a metal, a mixture including the metal, and a compound including the metal. For example, the hydrogen absorbing layer **180** may be configured to include a metal, a mixture of a metal, a compound of a metal, a metal and a mixture of the metal, a metal and a compound of the metal, a mixture of a metal and a compound of the metal, or a metal, a mixture of the metal and a compound of the metal. Here, the metal may include one or more of an alkali metal, an alkaline earth metal, a rare earth metal, a titanium (Ti) group metal, a transition metal, and a post-transition metal. Further, the metal may be particles having a diameter of several to several tens of nanometers (nm). For example, the metal may be particles having a diameter of 100 nm or less, but is not limited thereto.

[0079] The inorganic material of the hydrogen absorbing layer **180** can absorb hydrogen in various ways. For example, the inorganic material of the hydrogen absorbing layer **180** may absorb hydrogen through a chemical reaction or trap hydrogen in a gap between inorganic materials to absorb it.

[0080] The inorganic material of the hydrogen absorbing layer **180** reacts with the residual hydrogen in the color filter CF and the passivation layer **130** to produce a hydrogen compound. Accordingly, the hydrogen compound obtained by reacting the inorganic material of the hydrogen absorbing layer **180** with the residual hydrogen in the color filter CF and the passivation layer **130** may be included in the hydrogen absorbing layer **180**. For example, the hydrogen absorbing layer **180** may include the hydrogen compound as well as an inorganic material. The hydrogen compound generated in the hydrogen absorbing layer **180** may be formed through reaction as in the following [Reaction Formula 1].

$$\text{Me} + (\text{X}/2)\text{H}_2 \rightarrow \text{MeH}_x$$
 [Reaction Formula 1]

[0081] Reaction Formula 1 is a chemical reaction formula for explaining a process of producing a hydrogen compound (MeH_x) by reacting the inorganic material of the hydrogen absorbing layer **180** with the residual hydrogen. Here, Me means a metal. However, not only a metal, but also a metal compound or a metal mixture may react with hydrogen as described in [Reaction Formula 1] to thereby form a hydrogen compound.

[0082] Therefore, when the inorganic material of the hydrogen absorbing layer **180** reacts with the residual hydrogen in the color filter CF and the passivation layer **130** to produce a hydrogen compound, energy is generated or absorbed, and such energy can be called hydrogen adsorption energy.

[0083] In this regard, Table 1 describes the hydrogen adsorption energy according to a type of metal when the temperature is 298K. In connection with this, in a case in which the hydrogen adsorption energy has a positive number, it means that when a metal which is an inorganic material, and hydrogen are adsorbed, they react spontaneously while generating heat. In a case in which the hydrogen adsorption energy has a negative number, it means that when a metal which is an inorganic material, and hydrogen are adsorbed, they react involuntarily while absorbing heat.

TABLE-US-00001 TABLE 1 Hydrogen Adsorption Metal Type Energy [eV] Vanadium (V) −0.08 Nickel (Ni) 0.02 Palladium (Pd) 0.05 Copper (Cu) −0.05 Iron (Fe) −0.01

[0084] For example, since hydrogen adsorption reactions occur more spontaneously in the order of lower hydrogen adsorption energy, the hydrogen adsorption reactions occur better in the order of vanadium (V), copper (Cu), iron (Fe), nickel (Ni), and palladium (Pd).

[0085] However, in Table 1, the hydrogen adsorption energy when the temperature is 298K is described. Thus, the hydrogen adsorption energy may be further reduced as compared to those described in Table 1, since a process temperature is higher than 300K.

[0086] For example, the inorganic material included in the hydrogen absorbing layer **180** may include one or more of metals such as thorium (Th), zirconium (Zr), vanadium (V), palladium (Pd), titanium (Ti), magnesium (Mg), nickel (Ni), tin (Sn), platinum (Pt), chromium (Cr), silver (Ag), aluminum (Al), copper (Cu), gold (Au), cobalt (Co), iron (Fe) and the like, or lanthanum-nickel (La—Ni), lanthanum-nickel-aluminum (La—Ni—Al), nickel-magnesium (Ni—Mg), iron-titanium (Fe—Ti), and titanium-manganese (Ti—Mn), but is not limited thereto.

[0087] As the hydrogen absorbing filler, the organic material may include one or more of a hydroxy benzophenone-based compound, a hydroxy phenylbenzotriazole-based compound, and a hindered amines-based compound. For example, the organic material of the hydrogen absorbing layer **180** may be a HALS (hindered amine light stabilizer) represented by the following Chemical Formula 1, but is not limited thereto.

##STR00001##

[0088] The organic material can directly combine with hydrogen through a chemical reaction and absorb hydrogen.

[0089] The semiconductor layer A included in the light emitting display device according to an exemplary aspect of the present disclosure may be formed of an oxide semiconductor. The oxide semiconductor has excellent mobility and uniformity. For example, the oxide semiconductor may have a semiconductor layer formed of a quaternary metal oxide such as an indium tin gallium zinc oxide (InSnGaZnO)-based material, a ternary metal oxide such as an indium gallium zinc oxide (InGaZnO)-based material, an indium tin zinc oxide (InSnZnO)-based material, an indium aluminum zinc oxide (InAlZnO)-based material, an tin gallium zinc oxide (SnGaZnO)-based material, an aluminum gallium zinc oxide (AlGaZnO)-based material, and a tin aluminum zinc oxide (SnAlZnO)-based material, or a binary metal oxide such as an indium zinc oxide (InZnO)-based material, a tin zinc oxide (SnZnO)-based material, an aluminum zinc oxide (AlZnO)-based material, a zinc magnesium oxide (ZnMgO)-based material, a tin magnesium oxide (SnMgO)-based material, an indium magnesium oxide (InMgO)-based material, an indium gallium oxide (InGaO)-based material, an indium oxide (InO)-based material, a tin oxide (SnO)-based material, and a zinc oxide (ZnO)-based material. Composition ratios of respective elements included in the oxide semiconductor are not limited.

[0090] In this manner, when the semiconductor layer A is an oxide semiconductor, since the residual hydrogen included in the color filter CF and the passivation layer **130** diffuses to the semiconductor layer A composed of an oxide semiconductor, an electron concentration may increase due to reduction of the semiconductor layer A composed of an oxide semiconductor. As a result, a channel through which an unintended leakage current can flow may be formed in the semiconductor layer A formed of an oxide semiconductor, due to an increased electron concentration, so that a bright spot may be generated. Accordingly, the light emitting display device according to an exemplary aspect of the present disclosure removes the residual hydrogen included in the color filter CF and the passivation layer **130** to thereby solve problems that occur due to the semiconductor layer A composed of an oxide semiconductor. For example, even if hydrogen is diffused from the color filter CF and the passivation layer **130**, the diffused hydrogen is absorbed by the hydrogen absorbing filler of the hydrogen absorbing layer **180** and does not influence the semiconductor layer A composed of an oxide semiconductor. Therefore, the light emitting display device **100** can prevent a reduction in performance of the thin film transistor TFT due to residual hydrogen moving through the hydrogen absorbing layer **180**, reduce the occurrence of speckles and bright spots, and reduce uneven luminance.

[0091] The upper substrate **119** may be, for example, a metal, plastic film, and glass, but is not limited thereto. The metal may be, for example, iron-nickel (Fe—Ni), but is not limited thereto.

Therefore, even if a material of the upper substrate **119** is changed, defects such as bright spots and the like due to residual hydrogen can be reduced, so that the degree of freedom of the material of the upper substrate **119** can be improved.

[0092] The hydrogen absorbing filler in the hydrogen absorbing layer **180** may transmit light incident onto the hydrogen absorbing layer **180**. For example, the hydrogen absorbing layer **180** may have a transmittance of 50% or more with respect to light incident onto the hydrogen absorbing layer **180**. For example, the hydrogen absorbing layer **180** may have a transmittance of 70% or more with respect to light incident onto the hydrogen absorbing layer **180**. Light-transmissive characteristics of the hydrogen absorbing layer **180** may be advantageous in securing transmittance and improving luminous efficiency of a top emission type light emitting display device.

[0093] In some aspects, the hydrogen absorbing layer **180** may further include a getter, together with the hydrogen absorbing filler that absorbs hydrogen. The getter may be a particle that absorbs moisture and gas.

[0094] Further, in some aspects, the hydrogen absorbing filler that absorbs hydrogen may be included even in the overcoating layer **150** disposed on the thin film transistor TFT, and the bank **160** and the adhesive layer **170** disposed on the overcoating layer **150**.

[0095] The overcoating layer **150** and the bank **160** also include a hydrogen absorbing filler that absorbs hydrogen, so that the hydrogen absorbing filler included in the overcoating layer **150** and the bank **160** absorbs hydrogen through a chemical reaction, or hydrogen absorbing fillers trap hydrogen in a gap therebetween so that hydrogen can be absorbed.

[0096] Accordingly, residual hydrogen moving from the passivation layer **130** to the thin film transistor TFT can be more efficiently absorbed, and a degradation in performance of the thin film transistor TFT due to the moving residual hydrogen can be effectively prevented.

[0097] FIG. **3** is a cross-sectional view of one sub-pixel of a light emitting display device according to another exemplary aspect of the present disclosure. A light emitting display device **200** of FIG. **3** is different from the light emitting display device **100** of FIG. **2** in terms of an upper protective layer **290**, and other components thereof are substantially the same as those of the light emitting display device **100** of FIG. **2**. Thus, a duplicate description is omitted.

[0098] With reference to FIG. **3**, the upper protective layer **290** covers the light emitting element **140** to protect the light emitting element **140**. As illustrated in FIG. **2**, the upper protective layer **290** is conformally disposed on an upper surface of the cathode **143** of the light emitting element **140**. For example, the upper protective layer **290** may be formed along shapes of the upper surface and the side surface of the cathode **143**, but is not limited thereto. The upper protective layer **290** may extend to the upper surface and the side surface of the bank **160**.

[0099] In addition, various organic films or inorganic films may be used for the upper protective layer **290**, and the upper protective layer **290** may be formed of various structures such as an organic film-alone deposition structure, an inorganic film-alone deposition structure, or a structure in which an organic film and an inorganic film are alternately disposed. The upper protective layer **290** may be formed of, for example, silicon nitride (SiNx), silicon oxide (SiOx) or the like, but is not limited thereto.

[0100] Accordingly, in another exemplary aspect of the present disclosure, damage to the light emitting element **140** due to moisture or foreign matter can be prevented by further including the upper protective layer **290** for protecting the light emitting element **140**. Accordingly, element reliability of the light emitting display device **200** according to another exemplary aspect of the present disclosure may be significantly improved.

[0101] FIG. **4** is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure. A light emitting display device **300** of FIG. **4** is different from the light emitting display device **100** of FIG. **2** in terms of a disposition relationship of a light emitting element **340**, a bank **360**, and an adhesive layer **370**, and other

components thereof are substantially the same as those of the light emitting display device **100** of FIG. **2**. Thus, a duplicate description is omitted.

[0102] As illustrated in FIG. **4**, the bank **360** may extend between the light emitting area of the sub-pixel SP and the side end of the overcoating layer **150**. For example, a side end of the bank **360** may be disposed between a boundary of the light emitting area of the sub-pixel SP and the side end of the overcoating layer **150**. As illustrated in FIG. **4**, the side end of the bank **360** may be disposed more inwardly than the side end of the overcoating layer **150**.

[0103] In addition, a light emitting layer **342** and a cathode **343** may be disposed on the bank **360**. More specifically, the light emitting layer **342** may be disposed to extend on the bank **360**, and the cathode **343** may be disposed to cover a side surface of the light emitting layer **342** and an upper surface and a side surface of the bank **360**. However, as illustrated in FIG. **4**, the light emitting layer **342** is not limited to extending only to the upper surface of the bank **360**, but may also extend to the side surface of the bank **360**.

[0104] In addition, the adhesive layer **370** may be formed to cover the light emitting element **340**, the bank **360**, and the overcoating layer **150**. More specifically, the adhesive layer **370** may be disposed to cover an upper surface and a side surface of the cathode **343**, the upper surface and the side surface of the bank **360**, and only the upper surface of the overcoating layer **150**. For example, the adhesive layer **370** covers both an outer surface of the light emitting element **340** and an outer surface of the bank **360**, but may not cover all of an outer surface of the overcoating layer **150**.

[0105] In other words, a side end of the adhesive layer **370** may be disposed more inwardly than the side end of the overcoating layer **150**, but may be disposed more outwardly than the side end of the bank **360**. For example, the side end of the overcoating layer **150**, the side end of the adhesive layer **370**, and the side end of the bank **360** may be sequentially disposed based on an outside of the light emitting display device **300**. For example, since the overcoating layer **150** is disposed to cover all of the thin film transistor TFT, the gate insulating layer **120**, and the passivation layer **130**, adhesion between the overcoating layer **150** and the adhesive layer **170** may be improved as compared to adhesion between the gate insulating layer **120** and the passivation layer **130** and the adhesive layer **170**.

[0106] In addition, in the light emitting display device **100** according to another exemplary aspect of the present disclosure, the side end of the adhesive layer **370** may be disposed closer to the side end of the overcoating layer **150** than the side end of the bank **360**. For example, a distance between the side end of the adhesive layer **370** and the side end of the overcoating layer **150** may be shorter than a distance between the side end of the adhesive layer **370** and the side end of the bank **360**.

[0107] FIG. **5** is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure. A light emitting display device **400** of FIG. **5** is different from the light emitting display device **100** of FIG. **2** in terms of a disposition relationship of a light emitting element **440**, a bank **460**, and an adhesive layer **470**, and other components thereof are substantially the same as those of the light emitting display device **100** of FIG. **2**. Thus, a duplicate description is omitted.

[0108] As illustrated in FIG. **5**, the bank **460** may be disposed only in an area adjacent to the light emitting area of the sub-pixel SP. For example, a side end of the bank **460** may be disposed closer to the boundary of the light emitting area of the sub-pixel SP than the side end of the overcoating layer **150**. As illustrated in FIG. **5**, the side end of the bank **460** may be disposed more inwardly than the side end of the overcoating layer **150**.

[0109] In addition, a light emitting layer **442** and a cathode **443** may be disposed on the bank **460**. More specifically, the light emitting layer **442** may be disposed to extend on the bank **460**, and the cathode **443** may be disposed to cover an upper surface and a side surface of the light emitting layer **442**, and only an upper surface of the bank **460**. As illustrated in FIG. **2**, the light emitting layer **442** is not limited to extending only to the upper surface of the bank **460**, but may also extend

to a side surface of the bank **460**.

[0110] In addition, the adhesive layer **470** may be formed to cover the light emitting element **440**, the bank **460**, and the overcoating layer **150**. More specifically, the adhesive layer **470** may be disposed to cover an upper surface and a side surface of the cathode **443**, the upper surface and the side surface of the bank **460**, and only the upper surface of the overcoating layer **150**. For example, the adhesive layer **470** covers both an outer surface of the light emitting element **440** and an outer surface of the bank **460**, but may not cover all of the outer surface of the overcoating layer **150**.

[0111] In other words, a side end of the adhesive layer **470** may be disposed more inwardly than the side end of the overcoating layer **150**, but may be disposed more outwardly than the side end of the bank **460**. For example, the side end of the overcoating layer **150**, the side end of the adhesive layer **470**, and the side end of the bank **460** may be sequentially disposed based on an outside of the light emitting display device **400**.

[0112] In addition, in the light emitting display device **400** according to still another exemplary aspect of the present disclosure, the side end of the adhesive layer **470** may be disposed closer to the side end of the overcoating layer **150** than the side end of the bank **460**. For example, a distance between the side end of the adhesive layer **470** and the side end of the overcoating layer **150** may be shorter than a distance between the side end of the adhesive layer **470** and the side end of the bank **460**.

[0113] FIG. **6** is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure. A light emitting display device **500** of FIG. **6** is different from the light emitting display device **100** of FIG. **2** in that an adhesive layer **570** has a different structure without a separate hydrogen absorbing layer, and other components thereof are substantially the same as those of the light emitting display device **100** of FIG. **2**. Thus, a duplicate description is omitted.

[0114] As illustrated in FIG. **6**, the adhesive layer **570** is disposed between the light emitting element **140** and the upper substrate **119**. Specifically, the adhesive layer **570** may be formed to cover the light emitting element **140**, the bank **160**, and the overcoating layer **150** that are positioned thereunder, and may bond components disposed thereunder and the upper substrate **119** positioned thereon. In the light emitting display device **500** according to still another exemplary aspect of the present disclosure, as compared to the light emitting display device **100** of FIG. **2**, a separate hydrogen absorbing layer is not disposed on the adhesive layer **570**, and the upper substrate **119** is directly positioned on the adhesive layer **570**.

[0115] The adhesive layer **570** simultaneously includes an adhesive material and a hydrogen absorbing filler.

[0116] The adhesive material may be formed of an adhesive material in a liquid form or a film form, and may be formed of, for example, a resin, epoxy, or acrylic material, but is not limited thereto.

[0117] The hydrogen absorbing filler serves as a filter by absorbing or adsorbing hydrogen in order to absorb residual hydrogen in the color filter CF and/or the passivation layer **130**. The hydrogen absorbing filler may be an inorganic material or an organic material, and since it is substantially the same as the hydrogen absorbing filler described in FIG. **2**, a duplicate description thereof is omitted.

[0118] Since the adhesive layer **570** includes a hydrogen absorbing filler, residual hydrogen included in the color filter CF and the passivation layer **130** can be removed. That is, the adhesive layer **570** may simultaneously perform a function of bonding the lower substrate **110** and the upper substrate **119** and a function of removing hydrogen inside the light emitting display device **500**.

[0119] In the light emitting display device **500** according to another exemplary aspect of the present disclosure, a separate function of removing hydrogen inside the light emitting display device **500**, like the hydrogen absorbing layer, can be removed through the adhesive layer **570** that simultaneously performs an adhesion function and a hydrogen removal function. Thus, an overall

thickness of the light emitting display device can be reduced.

[0120] FIG. 7 is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure. A light emitting display device **600** of FIG. 7 is different from the light emitting display device **100** of FIG. 2 in terms of a configuration of an adhesive layer **670**, and other components thereof are substantially the same as those of the light emitting display device **100** of FIG. 2. Thus, a duplicate description is omitted.

[0121] As illustrated in FIG. 7, the adhesive layer **670** is disposed on the light emitting element **140**. The adhesive layer **670** may be formed to cover the light emitting element **140**, the bank **160**, and the overcoating layer **150** that are positioned thereunder, and may bond components thereunder and the upper substrate **119** positioned thereon.

[0122] The adhesive layer **670** simultaneously includes an adhesive material and an organic material **685** as a hydrogen absorbing filler.

[0123] Specifically, the adhesive material may be formed of an adhesive material in a liquid form or a film form, and may be formed of, for example, a resin, epoxy, or acrylic material, but is not limited thereto.

[0124] As the hydrogen absorbing filler, the organic material **685** of the adhesive layer **670** may include one or more of a hydroxy benzophenone-based compound, a hydroxy phenylbenzotriazole-based compound, and a hindered amines-based compound.

[0125] Since the adhesive layer **670** includes the hydrogen absorbing filler formed of the organic material **685**, the residual hydrogen included in the color filter CF and the passivation layer **130** can be removed. That is, the adhesive layer **670** may simultaneously perform the function of bonding the lower substrate **110** and the upper substrate **119** and the function of removing hydrogen inside the light emitting display device **600**.

[0126] A hydrogen absorbing layer **680** is disposed on the adhesive layer **670** in order to remove residual hydrogen included in the color filter CF and the passivation layer **130**. The hydrogen absorbing layer **680** includes a sealing material and an inorganic material as a hydrogen absorbing filler. With the exception that the hydrogen absorbing layer **680** in the light emitting display device **600** of FIG. 7 includes an inorganic material as a hydrogen absorbing filler, the hydrogen absorbing layer **680** is substantially the same as the hydrogen absorbing layer **180** described in FIG. 2. Thus, a duplicate description is omitted.

[0127] In the light emitting display device **600** according to another exemplary aspect of the present disclosure, the adhesive layer **670** includes the hydrogen absorbing filler of the organic material **685**, and the hydrogen absorbing layer **680** disposed on the adhesive layer **670** includes the hydrogen absorbing filler of an inorganic material.

[0128] By sequentially stacking the adhesive layer **670** including the hydrogen absorbing filler of the organic material **685** and the hydrogen absorbing layer **680** including the hydrogen absorbing filler of the inorganic material, the layers performing the hydrogen absorbing function can be implemented as a double layer, so that the residual hydrogen included in the color filter CF and the passivation layer **130** can be more easily removed.

[0129] In particular, when the hydrogen absorbing filler of the organic material **685** is included in the adhesive layer **670** positioned under the hydrogen absorbing layer **680**, the hydrogen absorbing filler of the organic material **685** has excellent dispersibility to an adhesive composition, so that it can be disposed on the light emitting element **140** without an aggregation phenomenon. When the hydrogen absorbing filler of the inorganic material is disposed adjacent to the thin film transistor TFT or the passivation layer **130**, optical properties of the light emitting display device may be degraded due to aggregation between the inorganic materials. In addition, since the inorganic material may damage the thin film transistor TFT or the passivation layer **130**, durability or light emission characteristics of the light emitting display device may be degraded.

[0130] Therefore, as in the light emitting display device of FIG. 7, by disposing the adhesive layer including the hydrogen absorbing filler of an organic material on the passivation layer and then,

disposing the hydrogen absorbing layer including the hydrogen absorbing filler of an inorganic material on the adhesive layer, the performance of removing hydrogen inside the light emitting display device may be improved without lowering durability or light emission characteristics of the light emitting display device.

[0131] FIG. 8 is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure. A light emitting display device **700** of FIG. 8 is different from the light emitting display device **600** of FIG. 7 in terms of a configuration of an adhesive layer **770**, and other components thereof are substantially the same as those of the light emitting display device **600** of FIG. 7. Thus, a duplicate description is omitted.

[0132] As illustrated in FIG. 8, the adhesive layer **770** is disposed on the light emitting element **140**. The adhesive layer **770** may be formed to cover the light emitting element **140**, the bank **160**, and the overcoating layer **150** that are disposed thereunder, and may bond components thereunder and the upper substrate **119** positioned thereon.

[0133] The adhesive layer **770** includes a first adhesive layer **771** and a second adhesive layer **772** disposed on the first adhesive layer **771**.

[0134] The first adhesive layer **771** may be formed to cover the light emitting element **140**, the bank **160**, and the overcoating layer **150** positioned thereunder. The first adhesive layer **771** simultaneously includes an adhesive material and a hydrogen absorbing filler of the organic material **685**. The second adhesive layer **772** is disposed on the first adhesive layer **771** and is formed of only an adhesive material without a hydrogen absorbing filler.

[0135] A hydrogen absorbing layer **780** including a hydrogen absorbing filler of an inorganic material and a sealing material is disposed on the adhesive layer **770**.

[0136] The light emitting display device **700** according to another exemplary aspect of the present disclosure has a structure in which the hydrogen absorbing filler of the organic material **685** is disposed only in a lower area of the adhesive layer **770**. That is, unlike the light emitting display device **600** of FIG. 7 having a structure in which the hydrogen absorbing filler of the organic material **685** is dispersed in the entire area of the adhesive layer **670**, the hydrogen absorbing filler of the organic material **685** is disposed only in the lower area of the adhesive layer **770**, that is, only in the first adhesive layer **771**, so as to be adjacent to the thin film transistor and the passivation layer **130**. The hydrogen absorbing filler of the organic material **685** is positioned close to the thin film transistor TFT, so that diffusion of hydrogen generated from moisture during a manufacturing process of a light emitting panel can be quickly controlled in a surface of the thin film transistor TFT.

[0137] FIG. 9 is a cross-sectional view of one sub-pixel of a light emitting display device according to still another exemplary aspect of the present disclosure. Since other components of a light emitting display device **800** of FIG. 9 are substantially the same as those of the light emitting display device **100** of FIG. 2, with the exception that an oxygen absorbing layer **880** instead of the hydrogen absorbing layer **180** is disposed. Thus, a duplicate description is omitted.

[0138] As shown in FIG. 9, in the light emitting display device **800** of FIG. 9, an oxygen absorbing layer **880** is disposed on the adhesive layer **170**. The oxygen absorbing layer **880** removes oxygen included in the light emitting display device **800**.

[0139] Specifically, when manufacturing the light emitting display device **800** by disposing the upper substrate **119** above the lower substrate **110**, external oxygen may be injected into the light emitting display device. In particular, external oxygen may move into the display device through a moisture-permeable prevention filler included in an organic material layer inside the light emitting display device used to prevent moisture permeation. Oxygen injected into the light emitting display device reacts with the organic material or the cathode constituting the light emitting element, and thus, a bright spot such as a white point or a dark point occurs, resulting in non-uniform luminance. Accordingly, defects of white spots can be solved by removing the oxygen inside the light emitting display device **800** through the oxygen absorbing layer **880**.

[0140] The oxygen absorbing layer **880** may include a sealing material and an oxygen absorbing filler **887**.

[0141] The sealing material may be formed based on a curable resin. For example, the curable resin may be a transparent thermosetting resin or a photocurable resin. For example, the curable resin may be an epoxy resin or an olefin resin.

[0142] At this time, the curable resin may have a polar group or a hydroxyl group. For example, the curable resin includes polyvinyl alcohol (PVA), polyvinylidene chloride (PVDC), ethylene vinyl alcohol (EVOH), polyacrylonitrile (PAN), polychlorotrifluoroethylene (PCTFE) or a combination thereof. In the case of a polymer having a polar group as a base material constituting the oxygen absorbing layer **880**, gas permeability is low and barrier property tends to be high, so that the amount of oxygen input from the outside to the inside of the light emitting display device **800** may decrease.

[0143] The oxygen absorbing filler **887** adsorbs or removes oxygen atoms or oxygen molecules through physical or chemical reactions. For example, the oxygen absorbing filler **887** may be an organic material or an inorganic material.

[0144] The oxygen absorbing filler **887** of the organic material may be carbon nano tube (CNT) or graphene, but is not limited thereto.

[0145] In addition, the oxygen absorbing filler **887** of the organic material may be an antioxidant such as a hindered phenyl-based compound, for example, IRGAFOS XP40, IRGAFOS XP60, IRGANOX 1010, IRGANOX 1035, IRGANOX 1076, IRGANOX 1135, or IRGANOX 1520L (which are trade names, manufactured by BASF), but is not limited thereto.

[0146] The oxygen absorbing filler **887** of the inorganic material may be any one or more of silica, zeolite, titania, zirconia, and montmorillonite, but is not limited thereto.

[0147] Meanwhile, the oxygen absorbing filler **887** of the inorganic material may be a surface-modified particle. For example, various coupling agents such as vinyl-based, glycidoxo-based, methacryl-based, amino-based, and mercapto-based silane coupling agents or alkoxide-based, chelating-based, and acylate-based titanium coupling agents may be used for a surface of a silica particle.

[0148] A surface treatment method of the silica particle may be performed by, for example, a stirring method, a wetting method, a drying method, or the like, but is not limited thereto.

[0149] Meanwhile, in the light emitting display device **800** of FIG. 9, the oxygen absorbing layer **880** may further include a hydrogen absorbing filler **885**. At this time, the oxygen absorbing layer **880** may include a sealing material, the oxygen absorbing filler **887** and the hydrogen absorbing filler **885**, absorb hydrogen remaining inside the light emitting display device **800**, and absorb or block oxygen injected from the outside of the light emitting display device **800**. In FIG. 9, although the oxygen absorbing layer **880** is illustrated to simultaneously include the oxygen absorbing filler **887** and the hydrogen absorbing filler **885**, the oxygen absorbing layer **880** may include only the oxygen absorbing filler **887** without the hydrogen absorbing filler as described above.

[0150] In the light emitting display device **800** of FIG. 9, a structure of the oxygen absorbing layer **880** may be substantially the same as the hydrogen absorbing layer **180** of the light emitting display device **100** of FIG. 2. Specifically, a side end of the oxygen absorbing layer **880** and the side end of the adhesive layer **170** may be collinearly disposed. Accordingly, the side end of the oxygen absorbing layer **880** may also be disposed more inwardly than the side end of the overcoating layer **150**, but may be disposed more outwardly than the side end of the bank **160**. For example, the side end of the overcoating layer **150**, the side ends of the adhesive layer **170** and the oxygen absorbing layer **880**, and the side end of the bank **160** may be sequentially disposed on an outside of the light emitting display device **800**. More specifically, the side end of the bank **160** may be disposed more inwardly than the side ends of the adhesive layer **170** and the oxygen absorbing layer **880**, and the side ends of the adhesive layer **170** and the oxygen absorbing layer **880** may be disposed more inwardly than the side end of the overcoating layer **150**.

[0151] Meanwhile, in the light emitting display device **800** of FIG. **9**, a structure in which the oxygen absorbing layer **880** is disposed on the adhesive layer **170** is illustrated, but is not limited thereto.

[0152] For example, the oxygen absorbing layer may be disposed under the adhesive layer **170**.

[0153] Also, the oxygen absorbing layer can be incorporated into the adhesive layer. Specifically, similar to a structure in which the hydrogen absorbing filler is dispersed inside the adhesive layer **170** in the light emitting display device **500** illustrated in FIG. **6**, oxygen absorbing fillers constituting the oxygen absorbing layer may be dispersedly disposed inside the adhesive layer **170**.

[0154] Also, similarly to a structure in which the hydrogen absorbing filler is disposed only in the first adhesive layer **771** which is a lower area of the adhesive layer **770** in the light emitting display device **500** illustrated in FIG. **8**, the oxygen absorbing filler constituting the oxygen absorbing layer may be disposed only in a lower area of the adhesive layer to be adjacent to the transistor and passivation layer.

[0155] The exemplary aspects of the present disclosure can also be described as follows:

[0156] According to an aspect of the present disclosure, there is provided a light emitting display device. The light emitting display device includes a lower substrate, a thin film transistor on the lower substrate, a passivation layer disposed on the thin film transistor and including hydrogen, an overcoating layer disposed on the passivation layer and planarizing the passivation layer, a light emitting element disposed on the overcoating layer and including an anode, a light emitting layer on the anode, and a cathode on the light emitting layer, a bank disposed on the overcoating layer and defining a light emitting area, an adhesive layer on the light emitting element and the bank, and a hydrogen absorbing layer disposed on the adhesive layer and including a hydrogen absorbing filler, wherein a side end of the bank is disposed more inwardly than side ends of the adhesive layer and the hydrogen absorbing layer, wherein the side ends of the adhesive layer and the hydrogen absorbing layer are disposed more inwardly than a side end of the overcoating layer.

[0157] The side end of the hydrogen absorbing layer and the side end of the adhesive layer may be collinearly disposed.

[0158] The side end of the bank may be disposed closer to the side end of the overcoating layer than a boundary of the light emitting area.

[0159] The side end of the bank may be disposed closer to a boundary of the light emitting area than the side end of the overcoating layer.

[0160] The light emitting layer may be disposed to cover an upper surface of the bank, and the cathode may be disposed to cover an upper surface and a side surface of the light emitting layer and to cover the upper surface of the bank.

[0161] The light emitting layer may be disposed to cover an upper surface of the bank, and the cathode may be disposed to cover an upper surface and a side surface of the light emitting layer and to cover the upper surface and a side surface of the bank.

[0162] The light emitting display device may further comprise an upper protective layer covering an upper surface of the light emitting element.

[0163] The adhesive layer may planarize the light emitting element and the bank.

[0164] The adhesive layer may be in contact with the overcoating layer, the light emitting element, and the bank.

[0165] The hydrogen absorbing filler may include one or more of an inorganic material and an organic material, and the inorganic material may include one or more of a metal, a mixture including the metal, and a compound including the metal, and the organic material includes one or more of a hydroxy benzophenone-based compound, a hydroxy phenylbenzotriazole-based compound, and a hindered amines-based compound.

[0166] The metal may include one or more of an alkali metal, an alkaline earth metal, a rare earth metal, a titanium (Ti) group metal, a transition metal, and a post-transition metal.

[0167] The metal may include one or more of thorium (Th), zirconium (Zr), vanadium (V),

palladium (Pd), titanium (Ti), magnesium (Mg), nickel (Ni), tin (Sn), platinum (Pt), chromium (Cr), silver (Ag), aluminum (Al), copper (Cu), gold (Au), cobalt (Co) and iron (Fe).

[0168] The mixture may include one or more of lanthanum-nickel (La—Ni), lanthanum-nickel-aluminum (La—Ni—Al), nickel-magnesium (Ni—Mg), iron-titanium (Fe—Ti), and titanium-manganese (Ti—Mn).

[0169] The metal may be a plurality of particles having a diameter of several to several tens of nanometers.

[0170] The hydrogen absorbing filler may be dispersed in the hydrogen absorbing layer.

[0171] The hydrogen absorbing layer may be included in the adhesive layer.

[0172] The hydrogen absorbing filler of the hydrogen absorbing layer may be a first hydrogen absorbing filler including the inorganic material, and the adhesive layer may further include a second hydrogen absorbing filler including the organic material.

[0173] The adhesive layer may include a first adhesive layer and a second adhesive layer on the first adhesive layer, and the hydrogen absorbing filler of the hydrogen absorbing layer may be a first hydrogen absorbing filler including the inorganic material, and the first adhesive layer may be formed of a second hydrogen absorbing filler including an adhesive material and the organic material, and the second adhesive layer may be formed of an adhesive material.

[0174] The thin film transistor may include a semiconductor layer, and the semiconductor layer may be formed of an oxide semiconductor or an amorphous semiconductor.

[0175] The anode may include a transparent conductive layer, or a reflective layer and a transparent conductive layer on the reflective layer.

[0176] The light emitting display device may further comprise a color filter disposed to overlap the light emitting element and including hydrogen.

[0177] One or more of the hydrogen absorbing layer and the adhesive layer may further include an oxygen absorbing filler.

[0178] The oxygen absorbing filler may include an inorganic material including one or more of an antioxidant, carbon nanotube (CNT), and graphene; or an organic material including one or more of silica, zeolite, titania, zirconia, and montmorillonite.

[0179] The oxygen absorbing filler of the organic material may be a particle surface-modified with vinyl-based, glycidioxy-based, methacryl-based, amino-based, and mercapto-based silane coupling agents or alkoxide-based, chelating-based, and acylate-based titanium coupling agents.

[0180] According to another aspect of the present disclosure, there is provided a light emitting display device. The light emitting display device comprises a lower substrate, a thin film transistor on the lower substrate, a passivation layer disposed on the thin film transistor and including hydrogen, an overcoating layer disposed on the passivation layer and planarizing the passivation layer, a light emitting element disposed on the overcoating layer and including an anode, a light emitting layer on the anode, and a cathode on the light emitting layer, a bank disposed on the overcoating layer and defining a light emitting area, and an adhesive layer disposed on the light emitting element and the bank and including a hydrogen absorbing filler or an oxygen absorbing filler. And, the hydrogen absorbing filler or the oxygen absorbing filler is disposed to be adjacent to the passivation layer.

[0181] The adhesive layer may include a first adhesive layer disposed on the passivation and a second adhesive layer disposed on the first adhesive layer, and the first adhesive layer may be composed of a curable resin and one or more of the hydrogen absorbing filler and the oxygen absorbing filler, and the second adhesive layer may be composed of a curable resin.

[0182] The hydrogen absorbing filler may include one or more of a hydroxy benzophenone-based compound, a hydroxy phenylbenzotriazole-based compound, and a hindered amines-based compound.

[0183] The oxygen absorbing filler may include an inorganic material including one or more of an antioxidant, carbon nanotube (CNT), and graphene; or an organic material including one or more of

silica, zeolite, titania, zirconia, and montmorillonite.

[0184] The curable resin of the first adhesive layer may include one or more of polyvinyl alcohol (PVA), polyvinylidene chloride (PVDC), ethylene vinyl alcohol (EVOH), polyacrylonitrile (PAN) and polychlorotrifluoroethylene (PCTFE).

[0185] The light emitting display device may further comprise a hydrogen absorbing layer disposed on the adhesive layer and including a hydrogen absorbing filler of an inorganic material including one or more of a metal, a mixture including the metal, and a compound including the metal.

[0186] Although the present disclosure is described in detail with reference to the accompanying drawings, the present disclosure is not limited thereto and may be embodied in many different forms without departing from the technical concept of the present disclosure. Therefore, the present disclosure are provided for illustrative purposes only but not intended to limit the technical concept of the present disclosure. The scope of the technical concept of the present disclosure is not limited thereto. Therefore, it should be understood that the above-described exemplary aspects are illustrative in all aspects and do not limit the present disclosure. The protective scope of the present disclosure should be construed based on the following claims, and all the technical concepts in the equivalent scope thereof should be construed as falling within the scope of the present disclosure.

Claims

1. A light emitting display device, comprising: a lower substrate; a light emitting element disposed on the lower substrate and including an anode, a light emitting layer on the anode, and a cathode on the light emitting layer; an adhesive layer including a first adhesive layer disposed on the light emitting element and a second adhesive layer disposed on the first adhesive layer, wherein the first adhesive layer includes a curable resin and one or more of a hydrogen absorbing filler and an oxygen absorbing filler.
2. The light emitting display device of claim 1, wherein the second adhesive layer includes a curable resin.
3. The light emitting display device of claim 1, further comprising: a thin film transistor disposed on the lower substrate; a passivation layer disposed on the thin film transistor; an overcoating layer disposed on the passivation layer and planarizing the passivation layer; and a bank disposed on the overcoating layer and defining a light emitting area.
4. The light emitting display device of claim 3, wherein the first adhesive layer covers an upper surface and a side surface of the bank and the cathode.
5. The light emitting display device of claim 3, wherein the bank has a side end disposed more inwardly than side ends of the adhesive layer.
6. The light emitting display device of claim 3, wherein the side end of the bank is disposed closer to the side end of the overcoating layer than a boundary of the light emitting area.
7. The light emitting display device of claim 3, wherein the side end of the bank is disposed closer to a boundary of the light emitting area than the side end of the overcoating layer.
8. The light emitting display device of claim 1, wherein the hydrogen absorbing filler includes one or more of a hydroxy benzophenone-based compound, a hydroxy phenylbenzotriazole-based compound, and a hindered amines-based compound.
9. The light emitting display device of claim 1, wherein the oxygen absorbing filler includes an organic material including one or more of an antioxidant, carbon nanotube (CNT), and graphene or an inorganic material including one or more of silica, zeolite, titania, zirconia, and montmorillonite.
10. The light emitting display device of claim 1, wherein the curable resin of the first adhesive layer includes one or more of polyvinyl alcohol (PVA), polyvinylidene chloride (PVDC), ethylene vinyl alcohol (EVOH), polyacrylonitrile (PAN) and polychlorotrifluoroethylene (PCTFE).
11. A light emitting display device comprising: a lower substrate; a light emitting element disposed on the lower substrate and including an anode, a light emitting layer on the anode, and a cathode on

the light emitting layer; an adhesive layer disposed on the light emitting element and including a hydrogen absorbing filler or an oxygen absorbing filler; and a hydrogen absorbing layer disposed on the adhesive layer and including a hydrogen absorbing filler of an inorganic material including one or more of a metal, a mixture including the metal, and a compound including the metal.

12. The light emitting display device of claim 11, further comprising: a thin film transistor disposed on the lower substrate; a passivation layer disposed on the thin film transistor; an overcoating layer disposed on the passivation layer and planarizing the passivation layer; a bank disposed on the overcoating layer and defining a light emitting area.
